Math is Beauty — Homework

Lesson 10-2 • Standard MPP.7

Name: _____ **Period:** _____ **Date:** _____

Today's Goals

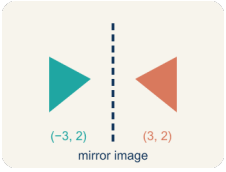
Content: I can use bilateral symmetry to describe structure in nature and in human creations.

Language: I can describe a symmetric design using the words mirror image, line of symmetry, and balance.

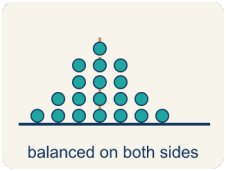
Key Words



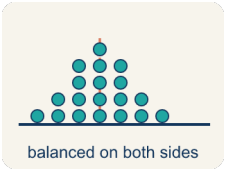
symmetric (simétrico) — Having two halves that match — one side is a mirror image of the other.



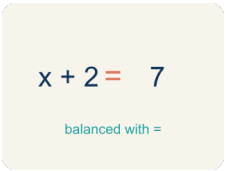
mirror image (imagen reflejada) — A copy of a figure flipped across a line, so the two halves face each other.



bilateral symmetry (simetría bilateral) — Reflective symmetry in plants and animals — the two sides of the body mirror each other.



line of symmetry (eje de simetría) — The line you can draw through a figure so each side is a reflection of the other.



balance (equilibrio) — An even, steady arrangement — neither side feels heavier than the other.

Practice

Show your work. Use the blank space under each problem.

1. What does it mean that a butterfly is symmetric?

A. The butterfly has exactly two wings
B. The two sides of the butterfly look like mirror images
C. The butterfly's wings are very colorful
D. The butterfly is larger than most insects

---

2. In plants and animals, reflective symmetry is called:

A. rotational symmetry
B. a mirror image
C. bilateral symmetry
D. balance

3. Why do people find symmetry pleasing?

A. Our brains are wired to favor balance, and bilateral symmetry provides balance
B. Symmetric things are always brightly colored
C. Symmetry only appears in rare and expensive objects
D. Symmetric shapes are always larger than non-symmetric ones

4. There are over 17,000 species of butterflies, and ALL butterflies have reflective symmetry. About how many of those species have reflective symmetry?

A. About half of them
B. Only the most colorful ones
C. None — symmetry is only in plants
D. All 17,000+ of them

5. Are the two leaves in a real plant layer EXACTLY symmetric? Why might this be?

6. A plant grows 2 leaves at a time, one layer per week. How many leaves has it grown after 6 weeks?

A. 6 leaves
B. 8 leaves
C. 12 leaves
D. 36 leaves

7. Where else do you see bilateral symmetry — in nature OR in something humans made?

8. Which is the best test for whether a figure has bilateral symmetry?

A. Check whether the figure has an even number of parts
B. Find a line where one side is the mirror image of the other
C. Check whether the figure is found in nature
D. Check whether the figure looks beautiful to you

9. The lesson says bilateral symmetry influences human creations 'from art to music to architecture.' Pick ONE of those three and describe how symmetry appears in it.

10. How many lines of symmetry does each figure have?

Figure	Think about the folds	Lines of symmetry
Square	2 through the sides + 2 through the corners	
Rectangle (not a square)	through the sides only — corner folds fail	
Equilateral triangle	one through each corner	
The letter H	one vertical + one horizontal	